Assignment .03

Network and Cyber Security (CSE900TR1)

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Q1. Case Study: Secure Communication Between Two Branch Offices

A multinational company is sending **sensitive financial data** over the internet. Since the internet is not secure, attackers can intercept or modify the data. Therefore, the company needs **network-layer security**.

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1. Which protocol suite should be used at the IP layer?

The correct protocol suite is **IPSec (Internet Protocol Security)**.

Detailed Explanation

IPSec is a collection of protocols designed to secure communication at the **Network Layer (Layer 3)** of the OSI model.

Unlike application-level security (like HTTPS), IPSec secures **all IP packets**, regardless of the application (email, file transfer, etc.).

Why IPSec is used:

- Protects **confidential data** from being read
- Ensures **data integrity** (no tampering)
- Verifies **sender authenticity**
- Works **transparently** (no need to modify applications)

☞ It is commonly used in:

- **VPNs (Virtual Private Networks)**
- Secure communication between branch offices

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2. Which two protocols provide authentication and encryption?

IPSec uses two main protocols:

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1. Authentication Header (AH)

Functions:

- Provides **authentication**
- Ensures **data integrity**
- Protects against **replay attacks**

How it works:

- Adds a header containing a **hash value**
- Receiver recalculates the hash
- If both match → data is trusted

✕ Limitation:

- Does **NOT encrypt data**
- Anyone can still read the content

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2. Encapsulating Security Payload (ESP)

Functions:

- Provides **encryption (confidentiality)**
- Provides **authentication**
- Ensures **data integrity**

Why ESP is more important:

ESP not only protects data from modification but also **hides the content**, making it the most widely used IPSec protocol.

Q2. Concept and Working of Encapsulating Security Payload (ESP) in IPSec

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Concept of ESP

Encapsulating Security Payload (ESP) is a protocol that protects data by **encrypting and encapsulating it inside a secure packet**.

☞ In simple terms:
ESP wraps your data in a **secure encrypted layer**, so even if someone intercepts it, they cannot understand it.

Structure of ESP Packet

ESP adds multiple fields to secure the data:

1. **ESP Header**
   - Security Parameters Index (SPI)
   - Sequence Number
2. **Payload Data**
   - Actual data (encrypted)
3. **Padding**
   - Used for alignment
4. **Authentication Data**
   - Used for integrity check

Working of ESP (Step-by-Step Explanation)

Step 1: Original Data Creation

- Sender creates a normal IP packet with sensitive data.

Step 2: Encapsulation

- ESP takes the original payload.
- It **encapsulates (wraps)** it inside a new structure.

Step 3: Encryption

- The payload is encrypted using algorithms like:
  - AES (Advanced Encryption Standard)
  - 3DES

☞ This ensures:

- No unauthorized user can read the data.

Step 4: Adding ESP Header

- ESP adds:
  - **SPI** → identifies security association
  - **Sequence Number** → prevents replay attacks

Step 5: Authentication

- A **hash (HMAC)** is generated.
- Ensures:
  - Data is not modified
  - Sender is verified

Step 6: Transmission

- The encrypted packet is sent over the internet.

Even if hackers intercept it:
☞ They only see encrypted data (useless without key)

Step 7: Reception and Verification

At the receiver side:

1. Check **sequence number** → prevents replay
2. Verify **authentication hash**
3. If valid → decrypt data
4. Retrieve original message

Modes of ESP

1. Transport Mode

- Only **data (payload)** is encrypted
- Original IP header remains visible

👉 Used for:

- End-to-end communication

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2. Tunnel Mode

- Entire IP packet is encrypted
- New IP header is added

👉 Used for:

- VPNs (branch-to-branch communication)

Advantages of ESP

- Strong **confidentiality (encryption)**
- Ensures **data integrity**
- Provides **authentication**
- Protects against **replay attacks**
- Widely used in **VPN security**

Real-Life Analogy

Think of ESP like:

🎁 Sending a **locked and sealed box**:

- Lock → Encryption
- Seal → Authentication
- Tracking number → Sequence number

Only the receiver with the correct key can open it.

✅ Final Conclusion

- IPSec secures communication at the IP layer
- AH provides authentication but no encryption
- ESP provides **complete security (encryption + authentication + integrity)**
- ESP is essential for secure communication between branch offices